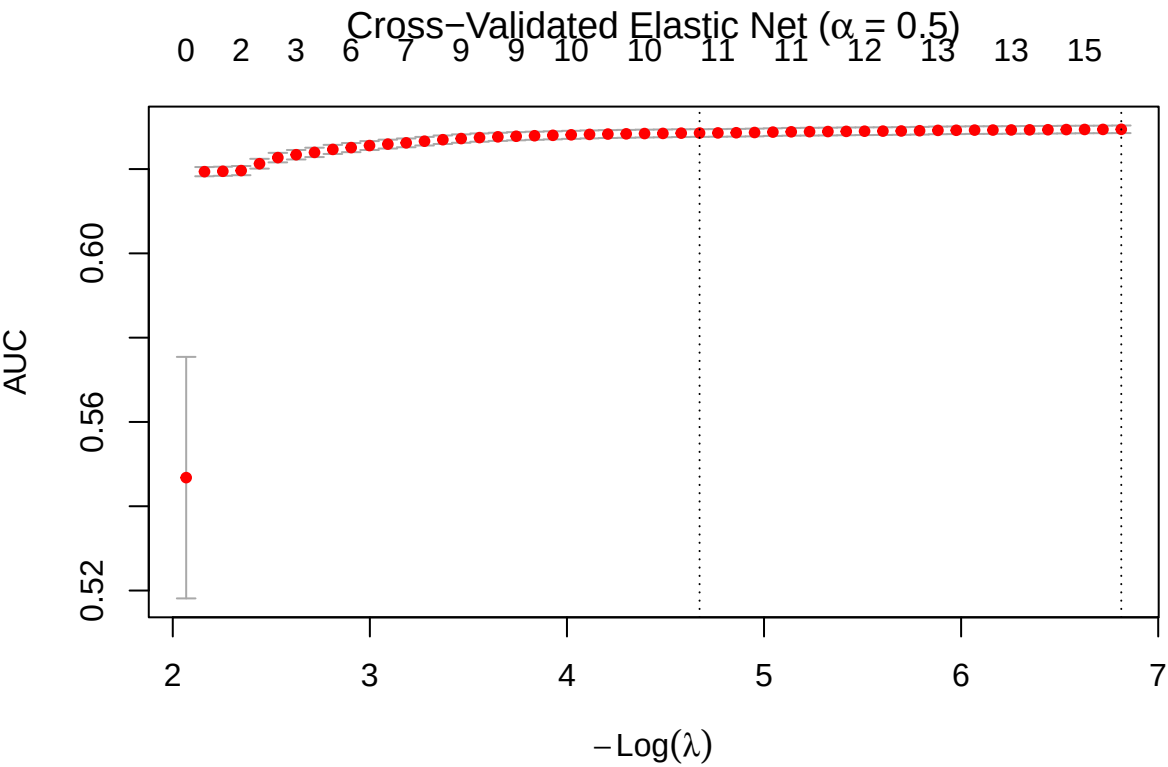


Elastic Model

2026-06-18

Contents



Optimal Lambda: 0.009344945

AUC: 0.6269

Optimal Threshold: 0.1946

Table 1: Elastic Net Binary Classification Performance

Metric	Value
Accuracy	0.5598
AUC	0.6269
F1 Score	0.3657
Matthews Correlation Coefficient	0.1536
Sensitivity / Recall	0.6585
Specificity	0.5362
Precision	0.2532
Balanced Accuracy	0.5974

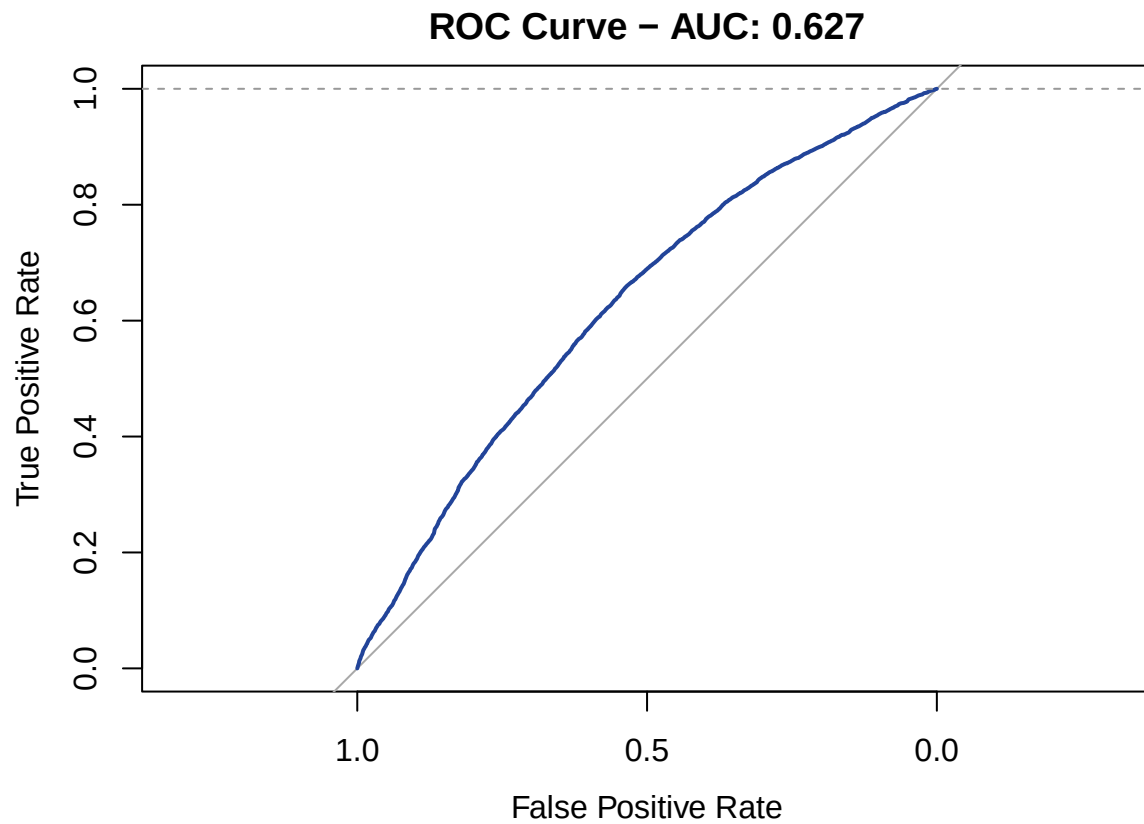
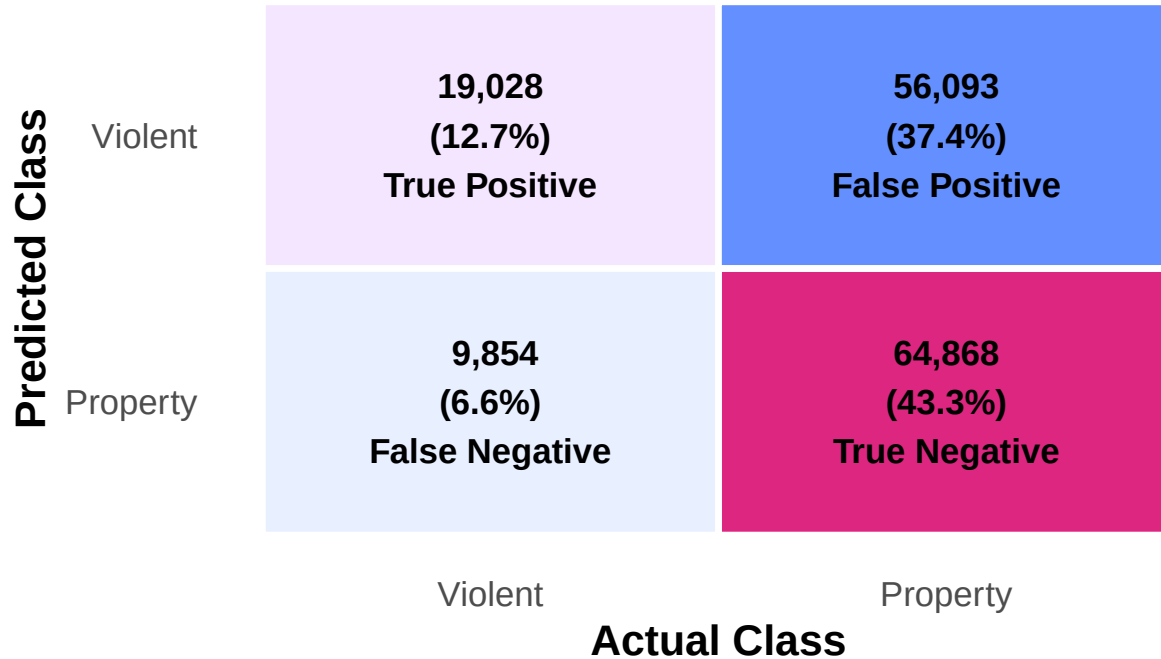


Table 2: Non-zero Elastic Net Coefficients

Predictor	Coefficient
edu_rate_college	-0.8396
single_parent_rate	0.7954
no_vehicle_rate	0.6839
edu_rate_no_hsdip	0.5177
disability_rate	0.2774
clusterLow Disadvantage	-0.1114
internet_rate	-0.0838
clusterHighest Disadvantage	0.0663
poverty_rate	0.0416

Confusion Matrix: Elastic Net

threshold = 0.195



Confusion Matrix and Statistics

##

##

	Reference						
## Prediction	Theft	Vehicle Theft	Theft	Vandalism	Burglary	Shoplifting	Assault
## Theft	3852		1212	1017	647	498	766
## Vehicle Theft	7405		3590	2287	1369	970	2138
## Vandalism	1263		441	430	212	143	341
## Burglary	13244		3668	2836	3900	1580	1650
## Shoplifting	12959		3265	2959	2490	3201	2150
## Assault	3236		1808	1263	541	491	1570
## Robbery	6329		2534	2182	1145	780	2627
## Sexual Offenses	803		305	233	152	192	196
## Other Violent	10178		6674	3825	1810	1041	5295

##

	Reference			
## Prediction	Robbery	Sexual Offenses	Other	Violent
## Theft	384		196	56
## Vehicle Theft	798		457	162
## Vandalism	172		84	22
## Burglary	846		497	138
## Shoplifting	1106		529	169
## Assault	711		282	143
## Robbery	1196		480	180
## Sexual Offenses	89		58	14
## Other Violent	2062		857	459

##

Overall Statistics

##

Accuracy : 0.1218

```

##          95% CI : (0.1202, 0.1235)
##    No Information Rate : 0.3955
##    P-Value [Acc > NIR] : 1
##
##          Kappa : 0.0376
##
##    McNemar's Test P-Value : <2e-16
##
## Statistics by Class:
##
##          Class: Theft Class: Vehicle Theft Class: Vandalism
## Sensitivity          0.06499          0.15279          0.02525
## Specificity          0.94727          0.87664          0.97984
## Pos Pred Value       0.44645          0.18721          0.13835
## Neg Pred Value       0.60756          0.84765          0.88685
## Prevalence           0.39555          0.15681          0.11367
## Detection Rate       0.02571          0.02396          0.00287
## Detection Prevalence 0.05758          0.12798          0.02074
## Balanced Accuracy     0.50613          0.51471          0.50254
##
##          Class: Burglary Class: Shoplifting Class: Assault
## Sensitivity          0.31795          0.35982          0.09383
## Specificity          0.82221          0.81818          0.93633
## Pos Pred Value       0.13752          0.11104          0.15630
## Neg Pred Value       0.93113          0.95294          0.89153
## Prevalence           0.08186          0.05937          0.11167
## Detection Rate       0.02603          0.02136          0.01048
## Detection Prevalence 0.18926          0.19239          0.06704
## Balanced Accuracy     0.57008          0.58900          0.51508
##
##          Class: Robbery Class: Sexual Offenses Class: Other Violent
## Sensitivity          0.162412          0.0168605          0.341772
## Specificity          0.885897          0.9864481          0.786245
## Pos Pred Value       0.068527          0.0284035          0.014254
## Neg Pred Value       0.953409          0.9771174          0.992485
## Prevalence           0.049146          0.0229578          0.008963
## Detection Rate       0.007982          0.0003871          0.003063
## Detection Prevalence 0.116478          0.0136279          0.214903
## Balanced Accuracy     0.524154          0.5016543          0.564008
##
## Per-Class F1:
##
##          Class: Theft Class: Vehicle Theft Class: Vandalism
##          0.1135          0.1683          0.0427
##          Class: Burglary Class: Shoplifting Class: Assault
##          0.1920          0.1697          0.1173
##          Class: Robbery Class: Sexual Offenses Class: Other Violent
##          0.0964          0.0212          0.0274
##
## Macro F1: 0.1054

```

Table 3: Non-zero Multinomial Elastic Net Coefficients by Class

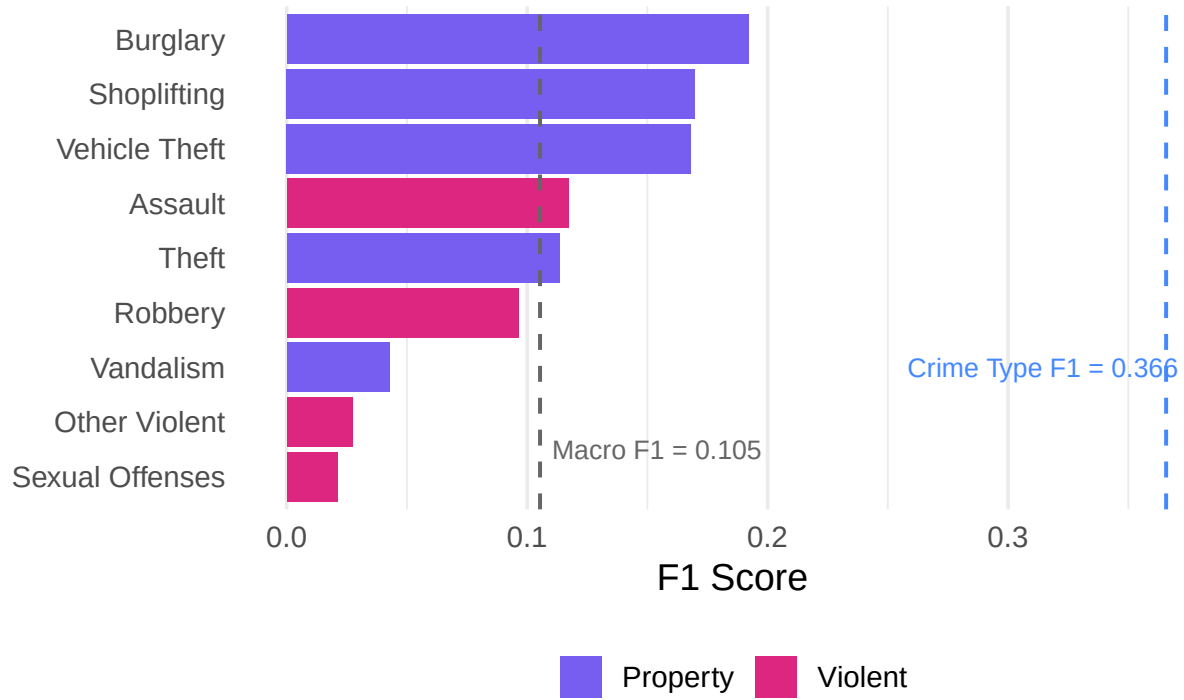
Class	Predictor	Coefficient
Assault	single_parent_rate	1.1544

Class	Predictor	Coefficient
Assault	no_vehicle_rate	0.5497
Assault	vacancy_rate	0.4988
Assault	disability_rate	0.4764
Assault	edu_rate_college	-0.4709
Assault	lang_isolation_rate	-0.3112
Assault	unemployment_rate	-0.1687
Assault	clusterLow Disadvantage	-0.0881
Assault	internet_rate	-0.0622
Assault	clusterModerate Disadvantage	-0.0115
Burglary	vacancy_rate	-1.1749
Burglary	single_parent_rate	-1.0624
Burglary	edu_rate_college	0.9435
Burglary	edu_rate_no_hsdip	-0.8002
Burglary	internet_rate	0.7445
Burglary	gini	0.5728
Burglary	renter_rate	-0.5130
Burglary	lang_isolation_rate	0.5093
Burglary	housing_burden_rate	0.4643
Burglary	unemployment_rate	0.2231
Burglary	clusterLow Disadvantage	0.2128
Burglary	clusterModerate Disadvantage	-0.0876
Burglary	clusterHighest Disadvantage	-0.0709
Burglary	no_vehicle_rate	-0.0707
Other Violent	single_parent_rate	1.0564
Other Violent	gini	0.8849
Other Violent	disability_rate	0.8591
Other Violent	edu_rate_college	-0.8342
Other Violent	housing_burden_rate	-0.7131
Other Violent	lang_isolation_rate	-0.4430
Other Violent	edu_rate_no_hsdip	0.3296
Other Violent	internet_rate	-0.3058
Other Violent	unemployment_rate	-0.1909
Other Violent	vacancy_rate	0.1203
Other Violent	clusterHighest Disadvantage	0.0829
Other Violent	poverty_rate	-0.0717
Other Violent	renter_rate	-0.0662
Robbery	no_vehicle_rate	0.9679
Robbery	housing_burden_rate	0.5288
Robbery	vacancy_rate	0.4105
Robbery	single_parent_rate	0.3547
Robbery	disability_rate	0.2887
Robbery	poverty_rate	-0.1687
Robbery	internet_rate	-0.1335
Robbery	clusterModerate Disadvantage	-0.1228
Robbery	renter_rate	0.0962
Robbery	clusterHighest Disadvantage	0.0465
Robbery	unemployment_rate	0.0400
Robbery	clusterLow Disadvantage	-0.0312
Robbery	edu_rate_college	-0.0205
Sexual Offenses	poverty_rate	0.7876
Sexual Offenses	single_parent_rate	-0.4925
Sexual Offenses	vacancy_rate	0.3993

Class	Predictor	Coefficient
Sexual Offenses	edu_rate_college	0.1988
Sexual Offenses	disability_rate	-0.1262
Sexual Offenses	lang_isolation_rate	0.1117
Sexual Offenses	gini	-0.0683
Sexual Offenses	clusterModerate Disadvantage	0.0101
Shoplifting	edu_rate_no_hsdip	-4.3723
Shoplifting	no_vehicle_rate	2.1620
Shoplifting	vacancy_rate	-1.0643
Shoplifting	unemployment_rate	0.9064
Shoplifting	gini	-0.8091
Shoplifting	renter_rate	0.7079
Shoplifting	housing_burden_rate	0.5828
Shoplifting	internet_rate	0.5283
Shoplifting	clusterLow Disadvantage	0.4887
Shoplifting	poverty_rate	0.3915
Shoplifting	disability_rate	-0.3082
Shoplifting	clusterModerate Disadvantage	0.2882
Shoplifting	single_parent_rate	-0.1188
Shoplifting	edu_rate_college	-0.0514
Shoplifting	clusterHighest Disadvantage	0.0333
Theft	edu_rate_college	1.1727
Theft	edu_rate_no_hsdip	-0.3145
Theft	gini	-0.1731
Theft	clusterHighest Disadvantage	-0.1340
Theft	no_vehicle_rate	-0.1169
Theft	clusterLow Disadvantage	0.0893
Theft	lang_isolation_rate	0.0412
Theft	single_parent_rate	-0.0161
Vandalism	gini	0.6370
Vandalism	unemployment_rate	-0.5082
Vandalism	edu_rate_college	0.3356
Vandalism	renter_rate	0.2749
Vandalism	housing_burden_rate	-0.2736
Vandalism	no_vehicle_rate	-0.1696
Vandalism	disability_rate	-0.1107
Vandalism	clusterHighest Disadvantage	-0.0788
Vandalism	clusterLow Disadvantage	0.0566
Vandalism	lang_isolation_rate	-0.0515
Vehicle Theft	no_vehicle_rate	-0.9407
Vehicle Theft	vacancy_rate	-0.3897
Vehicle Theft	gini	-0.2280
Vehicle Theft	disability_rate	-0.2060
Vehicle Theft	single_parent_rate	0.1458
Vehicle Theft	housing_burden_rate	-0.1456
Vehicle Theft	renter_rate	-0.0469
Vehicle Theft	clusterModerate Disadvantage	0.0245
Vehicle Theft	poverty_rate	-0.0129

Per-Subtype F1 Score

Gray dashed = subtype macro F1; blue dashed = crime type



Multinomial Elastic Net MCC: 0.0423

Table 4: Per-Class F1 and MCC — Multinomial Elastic Net

	Subtype	F1 Score	MCC
Burglary	Burglary	0.1920	0.0981
Shoplifting	Shoplifting	0.1697	0.1067
Vehicle Theft	Vehicle Theft	0.1683	0.0320
Assault	Assault	0.1173	0.0380
Theft	Theft	0.1135	0.0257
Robbery	Robbery	0.0964	0.0326
Vandalism	Vandalism	0.0427	0.0113
Other Violent	Other Violent	0.0274	0.0294
Sexual Offenses	Sexual Offenses	0.0212	0.0043